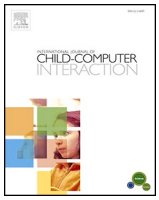
International Journal of Child-Computer Interaction 45 (2025) 100766



Contents lists available at ScienceDirect

International Journal of Child-Computer Interaction

journal homepage: www.elsevier.com/locate/ijcci



Research Paper



A semi-systematic literature review for improving assessment goals, practices, and strategies for promoting computational empowerment in K-12 education

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ARTICLE INFO

Keywords:
Child-computer interaction
Computational Empowerment
Learning
K-12 Education

ABSTRACT

Child-Computer Interaction (CCI) research has shown growing interest in Computational Empowerment (CE), yet there remains a lack of focus on assessment practices that can substantiate its outcomes. This paper presents a semi-systematic review to examine current approaches to assessing CE in CCI, investigate how these assessments connect to learning activities, and propose research directions for improving CE assessments. Our findings reveal that most studies are situated in formal educational settings and primarily aim to evaluate proposed technologies or learning models, but often with limited educator involvement. Assessment methods primarily rely on observations, feedback, reflection, and self-reports, but many do not specify theoretical frameworks. In response, we advocate for recentering participatory design in CE by fostering sustainable collaboration with educators. We propose strengthening CE assessment practices by building upon exemplary studies and focusing on three key areas aligned with the pillars of learning design: (1) refining and advancing CE learning goals, (2) integrating learning theories into assessment design, and (3) developing activities grounded in formative assessment principles.

1. Introduction

The field of Child-Computer Interaction (CCI) has increasingly focused on how children engage with digital technologies to build their functional, personal, and critical understandings of digital and emerging technologies (Van Mechelen et al., 2023). Within this agenda, the concept of Computational Empowerment (CE) has emerged as a strong research trajectory in CCI research. CE provides a framing that encompasses the methods by which students, both individually and in groups, develop the capacity to understand digital technologies, critically engage with its societal impacts, and participate in both the construction and deconstruction of these technologies (Dindler, Iversen, Caspersen, & Smith, 2022). As a relatively new concept, CE has been investigated from multiple perspectives within CCI research, yielding valuable empirical and theoretical contributions. These include studies on CE's foundational concepts in K-12 education (Dindler, Smith, & Iversen, 2020), its practical implications in teaching and learning (Göbl, Anna Guenther, Kayali, & Frauenberger, 2023; Schaper et al., 2022), its intersection with computational thinking and computing education (Odgaard, 2023; Lachney & Yadav, 2023; Musaeus, Van Mechelen, Caspersen, & Iversen, 2022; Iversen, Smith, & Dindler, 2018), its progression and future research directions (Smith et al., 2024; Kaspersen, Graungaard, Bouvin, Petersen, & Eriksson, 2021) as well as its connections to play (Tikkanen, Iivari, & Paananen, 2023), social justice (Iivari et al., 2023a; Iivari et al., 2023b), critical thinking (Morales-Navarro, Kafai, & Gregory, 2022; Tissenbaum, Sheldon, Seop, Lee, & Lao, 2017), and broader societal impact (Iivari et al., 2023b).

Despite these recent contributions, a notable gap exists in CE research: There is no systematic investigation of how CE is evaluated within educational contexts in CCI research. Although CE research is full of good intentions from CCI researchers, aimed at empowering the next generation in the digitized world, assessments to demonstrate impact are not in place. This lack has been highlighted as an important concern for the CCI community, as Van Mechelen and colleagues (Van Mechelen et al., 2023) articulated the need for advancing research efforts in assessment to further develop computer science and CE in education. Evaluations of CE have the potential to provide critical insights into the efficacy of CE-focused teaching strategies, highlighting areas for

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https://doi.org/10.1016/j.ijcci.2025.100766
Received 6 February 2025; Received in revised form 14 July 2025; Accepted 29 July 2025
Available online 29 July 2025
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refinement while fostering equitable learning opportunities for diverse student populations. Moreover, such evaluations can contribute to the development of more robust frameworks by aligning existing educational practices with the overarching goals of CE (Hjorth, Smith, Loi, Iversen, & Christensen, 2016; Van Mechelen et al., 2021b, 2023).

To address this gap, we conducted a semi-systematic literature review study (Snyder, 2019), aimed to investigate the following questions.

1. What are current assessment practices of CE in CCI research?
2. How are assessments connected to learning or research activities of CE?
3. What can be derived from CE research to further the development of sustainable and robust CE assessment frameworks?

Through this literature review, we identify the aspects of CE being learned and assessed, as well as the assessment methods, frameworks, and practices employed. The primary contributions of this study are threefold. First, we provide an overview of current CE assessment practices in CCI research, highlighting key themes and identifying gaps. Second, we synthesize how assessments are connected to learning or research activities of CE, exploring their implications for designing research and evaluating outcomes. Finally, we propose a future research agenda to integrate educators into CE research in order to advance CE practices. This agenda aims not only to support sustainable and robust CE assessment frameworks but also to inform related domains such as Computational Thinking and Design Literacy, where similar challenges around evaluating reflective and critical engagement persist.

2. Background

2.1. Computational empowerment

Empowerment refers to the process of providing individuals with knowledge, confidence, and opportunities that enable them to take ownership of their lives, make informed decisions and realize their full potential, ultimately enhancing their well-being and enabling meaningful contributions to society (Rappaport, 1981). As the society evolve into an increasingly digitized environment driven by rapid technological advancements, Iversen, Smith and Dindler (Iversen, Smith, & Dindler, 2018) introduced the concept of Computational Empowerment (CE) to K-12 computing education. The CE concept aims to foster children's agency in a digitized society through reflective and critical engagement with digital and emerging technologies, such as the internet of things (IoT), augmented reality (AR), and virtual reality (VR) (Van Mechelen et al., 2023). CE claims that empowerment is enabled through children's participation in both construction and deconstruction of technology (Dindler et al., 2022). CE has its root in the Participatory Design (PD) tradition drawing on participatory practices that support empowerment and democratic citizenship in a digital world (Dindler et al., 2020, 2022). Dindler and colleagues (Dindler et al., 2020) advocate for integrating PD approaches within computing education, noting that PD has a track record of adapting to technological advancements through learning, skill-development, and empowerment. Moreover, PD has a rich set of methods, practices and techniques that support co-designing technology and mutual learning (Dindler et al., 2020).

K-12 computing education has mainly been shaped by two overarching paradigms in CCI research: skill- and value-based approaches (Schulte et al., 2025). The skills-based approaches are largely centered around the concept of Computational Thinking (CT). CT (Wing, 2006) broadly refers to the ability to apply computing concepts to solve problems, design systems, and understand human behaviors across different fields. The concept of CT sparked widespread interests in CCI research (Giannakos, Papamitsiou, Markopoulos, Read, & Hourcade, 2020) and catalyzed global efforts to integrate CT into education (European Commission. Joint Research Centre, 2016). These efforts have led to the development of frameworks, tools and pedagogical practices for educating CT (Bers, González-González, & Ma Belén Armas–Torres, 2019; Carlborg, Tyrén, Heath, & Eriksson, 2019; Fagerlund, Vesisenaho, & Häkkinen, 2022) as well as CT-specific assessment frameworks (Allsop, 2019; Tang, Yin, Lin, Hadad, & Zhai, 2020). On the other hand, multiple strands of value-based approaches emerged in the CCI community (Van Mechelen, Musaeus, Iversen, Dindler, & Hjorth, 2021) and seek to foster children's critical engagement with technology and promote their digital citizenship from different perspectives (Dindler et al., 2022). These include critical perspectives (Lee, Choi, Lee, Choi, & Lee, 2023; Van Mechelen, Wagner, et al., 2021), Van Mechelen, Wagner, et al., 2021omputational participation (Kafai & Burke, 2013; Kafai, 2016), digital empowerment (Dindler, Iversen, Hjorth, Smith, & Nielsen, 2023; Kong, Wang, & Lai, 2019), critical literacy (Braswell, Johnson, Brown, & Payton, 2021), computational literacy (Kafai, Proctor, & Lui, 2019; Sosa et al., 2022), digital literacy (Bekker, Bakker, Douma, van der Poel, & Scheltenaar, 2015; Maqsood, Mekhail, & Chiasson, 2018), AI literacy (Druga, Christoph, & Ko, 2022; Long & Magerko, 2020), design literacy (Eriksson et al., 2019; Mahboob Kanafi, Iivari, Kinnula, & Sharma, 2022; Van Mechelen, Wagner, et al., 2021; Van Mechelen, Wagner, et al., 2021), computational action (Tissenbaum, Sheldon, & Abelson, 2019), computational perspective (Richard & Giri, 2019). These terms, while not explicitly framed as CE, address similar aims to promote children's empowerment in position to technologies. In line with the framing of these related concepts, this paper uses the terms competency and literacy interchangeably, as both emphasize not only technical skills but also critical awareness and agency. To clarify how CE relates to these established concepts, we provide an overview of each term's focus and its connection to CE. Table 1 summarizes their key emphases, illustrating how CE intersects with these approaches.

Despite fruitful research on CE focused on realizing these good intentions, significant gaps remain, particularly in assessment. While there are literature reviews on assessment of learning in CCI (Baykal, Eriksson, & Torgersson, 2023) and CT frameworks (Allsop, 2019), no comprehensive synthesis has focused on CE-specific assessments. A recent review (Van Mechelen et al., 2023) investigated how emerging technologies are taught in K-12 education, highlighting a lack of formative assessment and misalignment between learning objectives and activities. These limitations, as noted by Van Mechelen and colleagues (Van Mechelen et al., 2021b, 2023), pose challenges for teachers in effectively promoting CE. These challenges ultimately undermine children's opportunities to achieve CE competences, namely, their competences to critically reflect on the ethical and societal implications of technology through both the construction and deconstruction of technological systems. To advance CE beyond its good intentions requires a systematic approach to evaluating and consolidating CE assessment practices. This literature review addresses this gap by synthesizing current CE assessment efforts for proposing directions to further CE assessment in CCI. By bridging these gaps, this review seeks to support CE's mission to empower children as critical citizens in an increasingly complex and digitized world.

2.2. Learning and Assessment

Assessments of understanding and learning are central to achieving CE because they underpin children's ability to critically engage with and reflect on technologies. Thus, this section synthesizes literature on assessment and learning that we use to analyze the current CE assessment efforts. Learning is inferred through evidence derived from observations and assessments of learners' behaviors, expressions, and outputs (Schunk, 2012). Learning is a fundamental component in CE (Göbl et al., 2023), and the CCI community is increasingly reinforcing its connection to both CE research and learning (Eriksson, Baykal, & Torgersson, 2022). Given the CE's intrinsic relationship with learning, identifying effective assessment strategies is crucial for evaluating learning progress alongside developing learning content and delivery

Table 1
Overview of related-concepts and their connections to computational empowerment.

Concept	Key Focus/Definition	Connection to CE	Key References
Critical perspectives	Encourages critical thinking about data, AI, and digital systems, fostering reflective understanding of how technologies function and shape society	Enhances CE's critical dimension by promoting analytical awareness and reflection on the sociotechnical mechanisms and implications of digital technologies	(Lee et al., 2023; Van Mechelen, Wagner, et al., 2021)
Computational participation	Reframes computing education as socially and culturally situated, emphasizing learners' participation, identity, and community engagement in coding practices	Aligns with CE by emphasizing collective agency, sociocultural awareness, and participation as integral to meaningful engagement with digital technologies	(Kafai & Burke, 2013; Kafai, 2016)
Digital empowerment	Build individuals' digital competence and confidence, empowering them to participate meaningfully in digital society and positively influence their communities	Shares CE's emphasis on progressive empowerment and agency, highlighting digital skill acquisition as a foundation to meaningfully participate in and influence digital society	(Dindler et al., 2023; Kong et al., 2019)
Critical literacy	Cultivates awareness of how language and power operate within social, cultural, and political contexts, encouraging critical analysis and reflection	Deepens CE's critical dimension by encouraging learners to interrogate power structures embedded in digital tools, narratives, and sociotechnical systems	Braswell et al. (2021)
Computational literacy	Extend computational thinking skills to include personally meaningful design practices and critical examination of values and power in digital technologies	Fully aligns with CE's agenda by integrating technical fluency, design practices, and critical reflection to support empowered, meaningful digital participation	(Kafai, Proctor, & Lui, 2019; Sosa et al., 2022)
Digital literacy	The ability to use, understand, and evaluate digital technology and develop solutions to realize specific goals	Supports CE's foundation by fostering technical skills and critical understanding needed to critically navigate and shape the digital world	(Bekker et al., 2015; Maqsood et al., 2018)
AI literacy	Enable individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool in various contexts	Intersect with CE in fostering critical evaluation and agency, though its scope is domain-specific, focusing primarily on AI technologies	(Druga et al., 2022; Long & Magerko, 2020)
Design literacy	Engages learners in solving complex problems in a designerly way, emphasizing ethical awareness, decision-making, and the societal impact of technologies	Supports CE's emphasis on children's agency by fostering design competencies that empower them to take a proactive, protagonist role in technology	(Eriksson et al., 2019; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Van Mechelen, Wagner, et al., 2021)
Computational action	Emphasizes learning computing concepts as a tool to make meaningful, real-world impact, enabling youth to address issues in their communities	Fully aligns with CE's goal of fostering agency through technology by linking learning to socially impactful digital design practices to make meaningful impact	Tissenbaum et al. (2019)
Computational perspective	Describes how students relate to, value, and make personal meaning from computational practices in everyday life and learning contexts	Supports CE's emphasis on relevance and agency by fostering personal connections to computing that shape identity and critcal engagement with digital technology	Richard & Giri (2019)

methods (Fuentealba, 2011).

Schunk (Schunk, 2012) outlines various methods to provide learning evidence, summarized in Table 2, including "direct observation", "written responses", "oral responses", "rating by others", and "self--reports" (e.g., questionnaires, interviews and think-aloud protocols). These approaches, applicable to CE contexts, can provide more detailed insights into CE outcomes.

While there are other types and purposes of assessments, e.g., ipsative and authentic assessment (Eriksson, Kok, Barendregt, Jensen, & Nilsson, 2022), this literature review examines the scoped studies mainly through the lens of summative and formative assessment (Black & Wiliam, 1998; Wiliam, 2018), as it is mostly discussed and used among educators and in education research (Dolin, Black, Harlen, & Tiberghien, 2018; European Commission, 2018). Formative and summative assessments refer to different purposes of assessment: summative serves as measurement of learners' achievement or future potential, while formative assessment aimed at best support learning (Dolin et al., 2018). Summative assessments evaluate learning at the end of an instructional unit by benchmarking against standards and are suitable for reporting outcomes (Black & Wiliam, 1998). Conversely, formative assessment encompasses all the activities undertaken by teachers and students to provide feedback that informs the modification of teaching and learning activities (Black & Wiliam, 1998). Formative assessment focuses on learners' metacognition (knowing how to learn) and motivation (enjoying the learning process) (Wafubwa & Csíkos, 2022; Wiliam, 2018). Harlen and James (Harlen & James, 1997) argue that summative and formative assessments can complement each other, enabling educators to use them strategically to enhance student learning. Furthermore, formative assessment principles highlight connection between learning activities and learning goals (Wiliam, 2018). Similarly, constructive alignment coined by Biggs (Biggs, 1996) integrates constructivist learning theories into instructional design by aligning learning objectives, teaching activities, and assessments. It focuses on fostering high-level cognitive performances, ensuring learning activities and assessments systematically support the intended learning outcomes. Both CE and formative assessment highlight students' agency and ownership in the learning experiences and promote critical reflection and collaboration as key elements that influence their knowledge, skills, and actions.

As formative assessment is pertinent to CE promotion and learning activities, we analyzed not only summative assessment that occurs in the CE studies but also the connection between CE activities and formative assessment. We analyzed how the CE assessment practices align with these principles and suggest improvements to further integrate learning goals with assessment strategies to benefit both learners and the broader CCI community. Moreover, we will discuss potential research directions that could build upon existing assessment practices to construct more robust and effective CE assessment methods for educators and the CCI community.

Table 2
Methods of assessing learning, from Schunk [86:15].

Category	Definition
Direct observations	Instances of behavior that demonstrate learning
Written responses	Written performances on tests, quizzes, homework, papers
Oral responses	Verbalized questions, comments and responses during learning
Rating by others	Observers' judgements of learners on attributes indicative of learning
Self-reports	People's judgement of themselves: Questionnaires, interviews, stimulated recalls, think-aloud, dialogues

3. Methods

3.1. Literature search and selection

Following Snyder (Snyder, 2019), we selected a semi-systematic approach for this study because we wanted to generate an overview of the topic, map key themes, identify research gaps, and create an agenda for future research. For these reasons, our primary search focused on peer-reviewed, full research papers from the Association for Computing Machinery (ACM) Digital Library. Key venues include the Interaction Design and Children (IDC), CHI Conference on Human Factors in Computing Systems (CHI), FabLearn, International Computing Education Research (ICER), Designing Interactive Systems (DIS), Special Interest Group Computer Science Education (SIGCSE), Innovation and Technology in Computer Science Education (ITiCSE), Conference on Primary and Secondary Computing Education Research (WiPSCE), Participatory Design Conference (PDC), and International Computing Education Research (ICER). Additionally, we searched within journals such the International Journal of Child-Computer Interaction (IJCCI), given its foundational role in early CE research, and Transactions on Computing Education (TOCE).

To comprehensively capture CE-related efforts in CCI research, we broadened our search strategy beyond the term "computational empowerment" to include related subject-matter keywords reflecting similar research focus, as discussed in Section 2.1. This ensured the inclusion of terminologically varied but conceptually aligned work. The subject-matter keywords included: "computational empowerment", "design literacy", "critical literacy", "digital empowerment", "computational action", "digital literacy", "computational perspective", and "computational literacy". These subject-matter keywords were tested in search queries to validate their relevance and ensure the inclusion of critical CE papers, e.g., (Iivari et al., 2023b; Lachney & Yadav, 2023; Morales-Navarro, Kafai, & Gregory, 2022). This approach ensures that critical CE-related works using varied terminology are included in our corpus, broadening the scope of the review and capturing diverse contributions to this field.

Lastly, although we recognize CE's intersection with CT (Iversen, Smith, & Dindler, 2018), we excluded the term CT from our search due to the substantial existing literature on CT assessment in the CCI research, e.g., (Bers et al., 2019; Carlborg et al., 2019; Fagerlund et al., 2022; Tang et al., 2020). We combined assessment-associated keywords with each subject-matter keyword, to include all the variations of assessment-related concepts. These assessment-associated keywords include assess, evaluate, learning efficacy, progression, exam, test, learning outcome, measurement, and standard. We used the following search query across ACM full-text collection:

Full text: (subject-matter-keywords) AND Full text (assess*) OR Full text: (evaluat*) OR Full Text: ("learning efficacy") OR Full Text: (progression) OR Full Text: (exam) OR Full Text: (test) OR Full Text ("learning outcome") OR Full Text: (measurement) OR Full Text: (standard).

We conducted the search across ACM full-text collection based on the above-described systematic search rule. However, exceptions were made for the subject-matter keywords "*digital literacy*" and "*computational perspectives*", where the volume of irrelevant results necessitated limiting the search to the IDC venue rather than the ACM Full-Text Collection. As for IJCCI, all subject-matter keywords were searched without assessment-related terms, given the journal's focused scope on learning and children.

The search, conducted on 2024 May 22nd, yielded 303 publications (228 from ACM and 75 from IJCC). The combined records (n = 303) were saved in the Zotero reference management tool. Duplicate entries (n = 38) and non-full research papers (n = 7) were removed, resulting in 258 unique records for screening.

In the first screening, title and abstract were reviewed to exclude papers that (1) were literature reviews, or (2) mentioned the subject-matter-keywords only in passing without addressing, discussing, analyzing, or applying CE-related aspects. Papers with unclear eligibility were resolved through discussion with co-authors. A total of 192 records were excluded during this phase. For example, the study by Troiano and colleagues (Troiano et al., 2019) was excluded after discussion because it primarily focused on CT development and explored metrics for assessing CT through students-designed serious games but did not address other key aspects of CE. Another example of exclusion is the study by Mahboob and colleagues (Mahboob Kanafi, Iivari, & Kinnula, 2022), since this study does not aim to develop children's critical understanding and decision-making regarding digital technology and its societal impact. It is a PD paper that involved children but was not necessarily aligned with CE.

In the second screening, the first author conducted a full-text review of the remaining 66 papers. Papers were included if they meet at least one of the inclusion criteria: (1) explained methods for evaluating or inferring children's learning of CE aspects, (2) documented the development or assessment of CE-related learning activities, (3) described behavioral/task items aligned with CE aspects, (4) illustrated theoretical frameworks for CE assessment, and finally (5) analyzed learning activities promoting CE. Based on these inclusion criteria, 13 studies were excluded, leaving a final corpus of 53 papers (36 from ACM and 17 from IJCCI, see an overview in Table 3). The process is summarized in Fig. 1.

3.2. Data analysis

We conducted a thematic analysis based on the initial deductive categories derived from the research questions (Braun & Clarke, 2006;

Table 3
Overview of the final corpus with 53 included publications.

Year	ACM	IJCCI
2015–1 paper		(Bekker et al., 2015)
2016–2 papers	(Gourlet & Decortis, 2018; Hjorth et al., 2016)	
2017–2 papers	(Chu et al., 2017; Weibert et al., 2017)	
2018–4 papers	(Maqsood et al., 2018)	(Giannakos & Jaccheri, 2018; Gourlet & Decortis, 2018; de Paula, Burn, Noss, & Valente, 2018)
2019–6 papers	(Börjesson et al., 2019; Eriksson, Kok, Barendregt, Jensen, & Nilsson, 2022; Richard & Giri, 2019; Kafai, Proctor, & Lui, 2019; Kong et al., 2019; Zimmermann-Niefield et al., 2019)	
2020–4 papers	(Brooks & Sjöberg, 2020; Dowthwaite et al., 2020; Bilstrup & Kaspersen, 2020; Yap & Lee, 2020)	
2021–7 papers	(Antle et al., 2021; Braswell et al., 2021; Kaspersen et al., 2021; Lundmark & Jonsson, 2021; Van Mechelen, Wagner, et al., 2021)	(Hatzigianni et al., 2021; Kaspersen et al., 2021)
2022–13 papers	(Antle et al., 2022; Druga et al., 2022; Mahboob Kanafi, Iivari, & Kinnula, 2022; Morales-Navarro et al., 2022a, 2022b; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Vacca et al., 2022a, 2022b; Ventä-Olkkonen et al., 2022)	(Kaspersen et al., 2022; Schaper et al., 2022; Szczuka et al., 2022)
2023–10 papers	(Tamashiro et al., 2023; Lee et al., 2023; Mannila & Skog, 2023; Schaper et al., 2023)	(Odgaard, 2023; Chatlani et al., 2023; Dindler et al., 2023; Göbl et al., 2023; Iivari et al., 2023b; Xiang et al., 2023)
2024–4 papers	(Bustos et al., 2024; Dwivedi et al., 2024)	(Coenraad, 2024; Huang et al., 2024)
Total – 53 papers	36 papers from ACM	17 papers from IJCCI

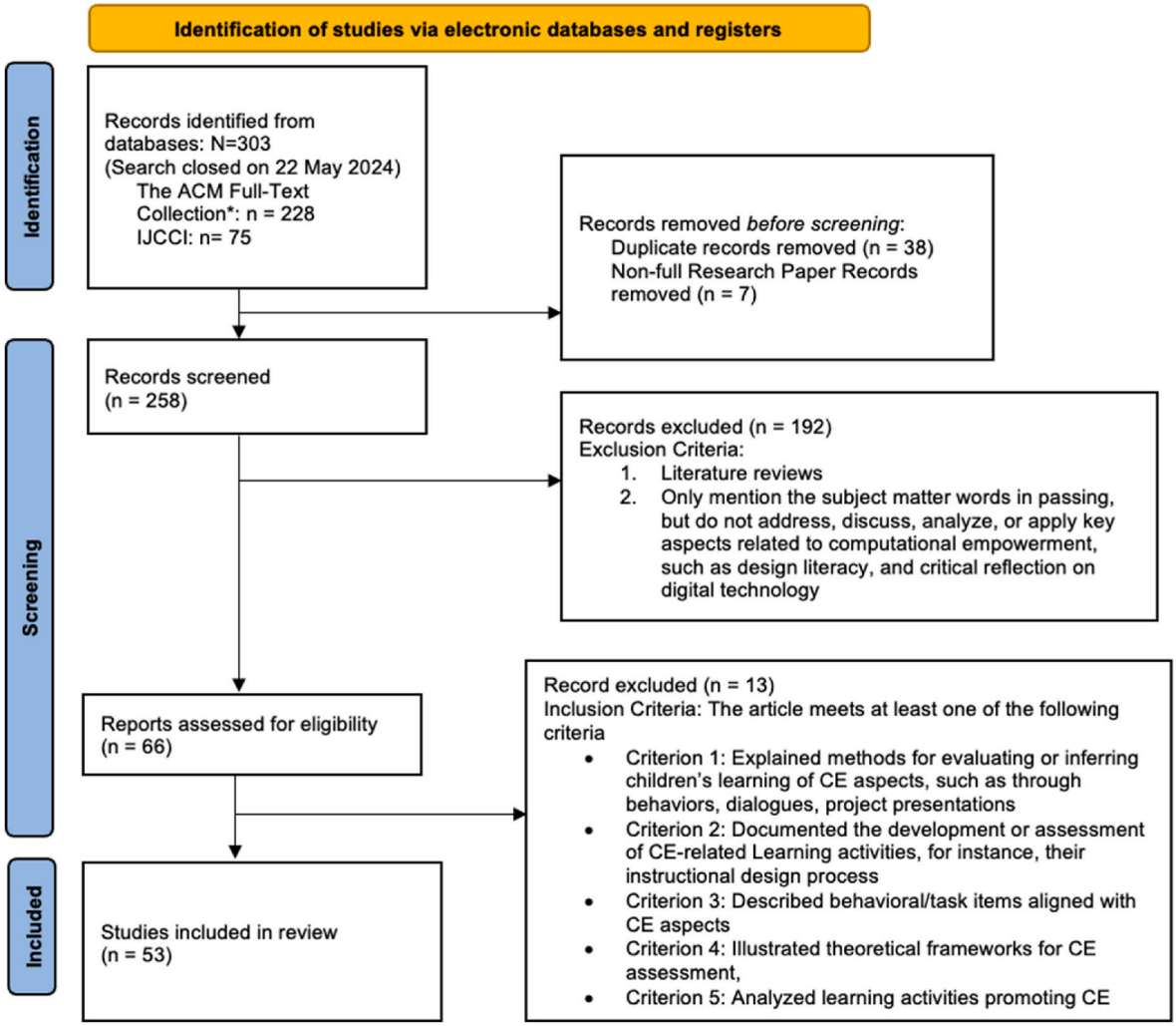


Fig. 1. Flow diagram based on the PRISMA-ScR guidelines (Page et al., 2021).

Snyder, 2019). Each of the 53 included records was read carefully and coded into the following deductive categories by the first author using NVivo. The deductive categories are "Educational Context," "Educational Level of Learners," "Roles of Stakeholders", "Purposes," "CE Aspects," "Learning Goals," "Learning Activities," "Theoretical Framework for Activity Development," "Assessment Methods," and "Assessment Theories" (See Table 4). All the textual components corresponding to the predefined categories were extracted and analyzed. During this process, new codes emerged inductively and were integrated into the coding scheme. The first author flagged records requiring for further discussion, and discrepancies were resolved collaboratively with the second and the third authors. To ensure consistency and validity, the open and iterative coding process was repeated across all 53 records. Categories were revisited to ensure saturation and the final coding scheme, including both deductive and inductive codes is presented in Table 4.

After coding, we identified patterns within and across child codes. For example, in the category "Assessment Methods," we adapted Schunk's (Schunk, 2012) categories of assessment methods (Table 2) in the initial data analysis. Yet, we grouped "project work" and "dialogues" as child codes under "observation," while omitting "written response" and "oral response" codes" due to the emphasis on hands-on, collaborative activities in CE learning contexts. We also introduced a new category, "Feedback and Reflection", with child codes for peer, teacher and self-feedback.

Next, the findings are summarized by category and reviewed by all authors listed in the authorship. We then compared findings across categories to identify major implications and future research directions. Finally, we reported detailed findings, illustrated with examples from the reviewed records, and proposed key areas for future research.

3.3. Limitations

We adopted a semi-systematic review approach (Snyder, 2019) to identify key themes and research gaps in CE assessment and to provide a focused analysis of relevant studies. However, we acknowledge the limitations inherent in this method. First, we do not claim exhaustive coverage of the topic in the literature, and our search strategy may have excluded pertinent studies, particularly due to biases in keyword selection and our concentrated focus on ACM venues (Snyder, 2019). This targeted scope may have led to an underrepresentation of studies published in the broader landscape of CCI research. As such, this review does not claim to comprehensively represent the entire CCI field.

Furthermore, the concepts central to CE, such as digital literacy, data literacy, and agency, are inherently multifaceted and dependent on contexts. To conduct a manageable review, we found it necessary to operationalize these constructs in narrower terms. By narrowing them, this study may obscure the broader complexity or overlook certain dimensions. Future research could further engage within the conceptual richness. Nevertheless, the selected corpus has been systematically analyzed to provide a robust foundation for understanding CE assessment practices within the scope of our research objective.

Finally, in this paper, we use the terms competency and literacy

Table 4
The Leading Questions for each Parent Code and Data-driven Child Codes.

Leading Questions	Deductive Parent Codes and Analysis Categories	Inductive Child Codes for Each Parent Code	Finding Section
RQ 1: What are current assessment practices of CE in CCI research?	Educational context Educational level of learners Roles of stakeholders Purposes	Formal/informal/ non-formal learning Pre-K/K1-6/K7-9/K9-12/K10-12 Roles of researchers, teachers, designers, students Evaluating learning activity Evaluating technology Evaluating theories Evaluating learning results	(Section 4.1) roles and purposes associated with CE assessments
	CE aspects Learning goals	Reflection/Digital literacy/Digital empowerment/ Design literacy With learning goals/ No alignment of learning goals and tasks/Progression/ Behavioral items	(Section 4.2) aspects of CE being assessed
	Assessment methods Assessment theories	Observation/ Feedback and reflection/Self-reports No theory specified/ CE literature/ Learning theory/HCI literature/ Sociocultural perspectives/ Multidisciplinary framework	(Section 4.3) Assessment methods and frameworks
RQ 2: How are assessments connected to learning or research activities of CE?	Learning activities Theoretical framework for activity development	Formative assessment alignment/ Experiential and interactive learning/ Structured reflection/ Introductory explanation CE and CT literature/ Learning theories	(Section 4.4) Connection between assessment and CE activities
RQ 3: What can be derived from CE research to further sustainable and robust assessment frameworks?	–	–	(Section 5) synthesized in the discussion section

interchangeably when referring to the ability to critically and effectively engage with digital systems based on the following reasons. While competency typically refers to demonstrated ability or performance, literacy often emphasizes not just technical skills but also understanding and interpretation within sociocultural contexts (e.g. (Gilster, 1997),). However, the boundaries between these terms are mostly blurred in the context of CE and the terms are often used interchangeably in CCI and computer science education, e.g., (Braun & Huwer, 2022; Kafai & Burke, 2014; Voogt, Fisser, Good, Mishra, & Yadav, 2015). Based on this overlap in usage and conceptual scope, we adopt an interchangeable use of the terms throughout this paper.

4. Findings

Our findings are organized in five sections. Sections 4.1-4.3 address RQ1: "What are current assessment practices of CE in CCI research?" and examine the roles, purposes, and methods associated with CE assessments. Section 4.4 addresses RQ2, "How are assessments connected to learning or research activities of CE?" and examines the diverse characteristics and theoretical frameworks that shape learning or research activities in the contexts of CE. For transparency and traceability, Appendix A.1 summarizes each finding with corresponding citations from the reviewed studies.

4.1. Roles and purposes associated with CE assessments

4.1.1. Roles, contexts, geographical location

Findings indicate that the development of CE assessments and activities in CCI research is predominantly driven by researchers. While other stakeholders, such as teachers and students, occasionally contribute, researchers remain the central actors in this development process. A majority of the reviewed papers (n = 25) rely exclusively on researchers to develop assessment and learning activities. While researchers are always involved in the development process, other stakeholders are teachers (n = 13), students (n = 6), designers/artists (n = 3), and multiple stakeholders, such as educational scientists, policymakers and publishers (n = 1). In addition, one paper reported on teacher-led observations of children's learning, where teachers executed the instruction independently without researcher involvement.

Shifting the focus to the educational contexts where these research efforts take place, here is an overview of the formal, non-formal and informal learning environments represented in the reviewed studies. The most common learning context is formal learning settings within school systems (n = 29), followed by non-formal learning settings (n = 14), such as after-school programs, community programs, and labs. Four papers address both formal and non-formal learning contexts, while four papers focus on informal learning contexts, such as family settings and unstructured learning environments. Two papers broadly address the learning of CE, which can be applied to all learning contexts. The educational levels targeted by the studies in the scoped review are: grades K-6 (n = 26), 7–9 (n = 21), K10-12 (n = 11), K-12 in general (n = 6), kindergarten (n = 4), and youth (aged 15–24, n = 1). These findings highlight a discrepancy: although CE research primarily targets formal educational contexts, especially from K-9, the involvement of education practitioners is limited. This suggests that CE has yet to be widely embraced or implemented in educational settings by frontline educators.

The geographic distribution of studies reveals a strong regional focus, with North America and Scandinavia leading CE research that involves evaluation and assessment. Most studies have been conducted in North America (USA: n = 13, Canada: n = 3, USA and Canada: n = 1) and Europe (n = 30), with the highest representation in Scandinavian countries: Denmark (n = 13), Finland (n = 4), and Sweden (n = 3). There are also two papers from China and one paper each from Korea, New Zealand, and Singapore. Additionally, one paper conducted by researchers in the US (Morales-Navarro, Kafai, & Gregory, 2022), focuses on a global learning project. This result reflects the historical prominence of participatory design and progressive educational practices in these regions, while other parts of the world remain underrepresented. It is possible that we have overlooked different notions from the Global South that could align with or contribute to the ongoing CE discourse. There is a risk that similar ideas are explored under different terminologies in the Global South, which we may have missed in this literature review and have not been discussed in CE literature.

This analysis illustrates that while CE research predominantly targets formal educational contexts and relies mainly on researchers for development, it remains geographically concentrated in the Global North and lacks broader engagement with education practitioners. The findings point to a need for more participatory and inclusive global distributed efforts to integrate CE into educational practices.

4.1.2. Purposes

We grouped findings into five thematic purposes: (1) evaluating the

effectiveness of learning models, (2) assessing technological designs for CE, (3) validating proposed CE frameworks, (4) deepening understanding of CE learning processes, and (5) research through design. Each purpose reflects distinct approaches to leveraging assessment in CE research. The majority of the reviewed studies papers (n = 19) employ assessment to evaluate the effectiveness of learning activity models designed to promote CE competencies. These assessments aim to measure how well learning interventions, such as workshops or classroom lessons, to help children develop CT and CE competencies. By measuring participant engagement, skill development and understanding, researchers illustrate their learning intervention outcome and refine their model designs to better foster CE. For instance, Dwivedi and colleagues (Dwivedi et al., 2024) proposed a model of participatory design activities for engaging children in AI problem formulation to foster children's AI literacies. Their empirical study aimed to provide empirical evidence to prove the effectiveness of their proposed learning activity model which incorporates co-design approaches with problem reduction heuristics in the context of teachable machines. They conclude the paper by discussing how their research results could inform the design of future participatory AI activities.

A second set of studies (n = 10) focuses on evaluating the learning efficacy of technological tools they designed to support CE. In these cases, assessment validates the design of the technology as a meaningful vehicle for fostering CE learning. For instance, Zimmermann-Niefield and colleagues (Zimmermann-Niefield et al., 2019) illustrated how they designed tools and experiences for children to learn machine learning within sports contexts. They proposed a mobile application, called AlpacaML, as a design example. Children could collect data through a wearable sensor to train the app in an iterative design process. Their empirical study illustrated the learning process and showcased results to prove the effectiveness of their design in achieving CE results, in this case, critical literacy.

A third set of studies (n = 9) uses assessment to evaluate the robustness and applicability of proposed CE frameworks in educational contexts. These papers synthesized existing practices and theories to propose new frameworks for CE. They empirically test their frameworks through assessments in empirical environments or incorporate the proposed CE framework into case studies. For example, Kaspersen and colleagues (Kaspersen et al., 2021) proposed a progression model of CE and assessed children's learning to test this progression model in real-world learning settings.

Another set of studies (n = 9) used assessment to better understand how children learn and make sense of CE-related topics. These papers use assessment to reveal how children conceptualize CE ideas and solve problems within learning activities, aiming to deepen our understanding of the learning process and explore cognitive and developmental processes and outcomes. For instance, Vacca and colleagues (Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022) investigates what kind of data reasoning and social-emotional students engage through creation of data comics. They then co-designed with teachers to create learning experience where student use digital storytelling tools to create comics around data. To extract learning evidence, researchers rely on the learning outputs, namely data comic artifacts, as well as interviews with students and teachers. Through these data, they conclude types of data reasoning and associated social-emotional reasoning that were apparent within the comic narratives, including "Describing", "Narratives to Contextualize the Data", "Situating the Self in Data", "Going Beyond Data".

Lastly, four papers followed a research-through-design approach, where assessment was used to iteratively develop ideas for new learning tools and mediums. For example, Yap and Lee (Yap & Lee, 2020) focused on creating an engaging medium for early adolescents to learn about managing their informational privacy and making informed decisions. Assessment in this context played a critical role in refining the design of learning tools, ensuring that the design not only catered to learners' interests but effectively imparted essential CE considerations and knowledge.

This analysis reveals that the purposes of CE assessments in CCI research are multifaceted, yet, the primary function is evaluative, focusing on demonstrating learning evidence and validating the effectiveness of learning models or technological designs and frameworks proposed by researchers. However, other aims of studies, such as exploring children's learning process of CE, also emerge in the review, pointing to opportunities for creating a more comprehensive and educational understanding of CE.

4.2. Aspects of CE being assessed

The reviewed studies incorporated aspects of CE into their learning topics, whether by specifying the CE aspects in their stated learning goals or by addressing them during learning activities without formally identifying them as learning goals. Themes of CE aspects emerged through analysis and are described as follows. Some studies addressed multiple CE aspects, resulting in a cumulative theme count that exceeds the total number of reviewed studies.

One predominant CE focus is "**critical reflection**" on the roles of technology (n = 27, 50 %), including considerations of societal impact, ethical issues, and technological and algorithmic bias. These studies emphasize the importance of student reflection through their hands-on activities and creativity in the learning experience. This aspect has been coupled with other CE aspects in the studies to reinforce the core CE value of critical reflection. Another prominent theme is "**digital literacy**" (n = 18, 34 %), which encompasses AI literacy, data literacy, and technical literacy. Digital literacy refers to the skills and competencies required to navigate the digital world, including the ability to create and consume media. With the growing integration of AI and data in society, these studies stress the importance of AI and data literacy to equip children with the skills and knowledge to make informed decisions about their engagement with these new technologies. "**Digital empowerment**" is another frequently addressed aspect (n = 18, 34 %). These papers focus on agency, highlighting the individual's ability to effectively use digital technologies to develop essential life skills, influence digital technology development and critically understand how digital technologies shape personal experiences and broader societal dynamics.

Finally, "**digital design literacy**" (Christensen, 2019), which encompasses making literacy, was addressed in 19 papers (36 %). They frame "digital design literacy" as the ability to appropriately use design and digital competences to create solutions tailored to specific contexts, purposes, and functions. These studies highlight the importance of educating children through a design-thinking process to critically solve problems, which shall equip them to tackle complex, real-world problems in their daily lives. Consequently, these papers advocate for learning through hands-on activities that embed design thinking processes directly into the learning experience, promoting active engagement and practical problem-solving skills.

The analysis reveals that the primary focus of CE as a learning topic is "critical reflection" on the roles of technology, which is usually coupled with other CE aspects such as digital literacy, digital empowerment and digital design literacy. These aspects are often incorporated into activities without explicit articulation as learning goals. To further explore this phenomenon, we also examine the articulation of learning goals and their progression in the reviewed studies.

4.2.1. Learning goals and progression

The reviewed studies exhibit varying degrees of clarity in articulating learning goals related to CE. While CE aspects are often integrated into activities, 26 papers do not specify explicit learning goals and their relation to CE aspects, as most studies' purposes as mentioned above focused more on validating tools than learning. In contrast, 18 papers outline learning goals, focused on broad objectives for interventions or workshops. For instance, two papers (Hatzigianni et al., 2021; Iivari

et al., 2023b) identified specific behavioral items, linking learning tasks and actions to targeted CE competences. These clear connections between learning actions and targeted CE competences provide valuable insights into constructive alignment for developing both summative and formative assessments. In addition, six papers addressed the concept of progression in CE skills. Three of these papers give examples of learning progression among different ages of pupils (Hatzigianni et al., 2021; Iivari et al., 2023b; Mannila & Skog, 2023). For example, Iivari and colleagues (Iivari et al., 2023b) provide learning goals and age-appropriate technology features in their learning intervention, respectively for school pupils aged 11–12 years old and 14–15 years old. The remaining three papers explore progression in the complexity of skills and understanding. For example, Schaper and colleagues (Schaper et al., 2022) introduce a three-tiered mastery framework to code the students' learning, ranging from understanding basic technology mechanism and its intended use (Level 1) to critically assessing societal implications of technologies (Level 3). Similarly, Kaspersen and colleagues (Kaspersen et al., 2021) employed the SOLO taxonomy to propose a sequential progression model for CE learning regarding the increasing complexity in learners' understanding of subjects. Finally, Vacca and colleagues (Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022) show a rubric to evaluate pupils' performance and understanding with examples of signs of learning. These examples illustrate the potential for developing structured and progressive learning goals in CE. They highlight the possibilities and the need for further research to align CE aspects with explicit learning goals, fostering deeper and more measurable CE outcomes.

4.3. Assessment methods and frameworks

We classified the methods into three main themes: observation, feedback and reflection, and self-reports. Most studies use multiple methods to demonstrate learning evidence. In regards with the assessment framework, most of the studies do not specify any theoretical frameworks they use to analyze or assess the learning outcomes. The rest of papers rely on theoretical foundation on Making, CT and CE literature, sociocultural perspectives, HCI literature, learning sciences and multidisciplinary theories, to develop assessment.

4.3.1. Assessment methods

The result shows that the assessment methods for CE rely heavily on observation (n = 41, 77 %), including observation of actions, dialogues, and tangible outputs. Observational methods are often complemented by self-reporting measures, such as interviews, questionnaires, and post- and pre-learning unit tests, at the end of the learning sessions, aligned with summative assessment purposes. Additional assessment methods include feedback mechanisms, such as peer feedback, self-feedback and teacher feedback, resonate with formative assessment principles. It is worth noting that some papers employ multiple methods, which is why the count for each method adds up to more than the total number of the papers (for an overview, see Appendix A.2).

4.3.1.1. Observation (n = 41, 77 %). The most prevalent assessment method in the reviewed studies is observation, mentioned in 41 papers as the primary tool for evaluating children's learning results. Observations can be very different in nature, in terms of what they observe, and how it demonstrates learning process and learning results. Researchers observed a variety of element to elicit learning evidence, including children's actions (behaviors and movements) (n = 29), dialogues (such as discussions and self-talk) (n = 28), and tangible outputs (such as digital artifacts, screenshots and physical objects) (n = 28). For instance, the study by Richard and Giri (Richard & Giri, 2019) documented dialogue and interaction among children, adults, and robots in a kindergarten to demonstrate how children learn CT and CE. This study highlighted adult-scaffolded activities with aligned learning goals, as well as instances of children independently using robots creatively, fostering empowerment without direct adult intervention. The study also emphasized children's ability to transform these interactions into more social, peer-driven experiences. In contrast, Zimmermann-Niefield and colleagues (Zimmermann-Niefield et al., 2019) used students' project work as learning evidence, such as a softball child player using AlpacaML to model good and bad pitching techniques. The design process enabled the child player to practice based on real-time feedback, highlighting how students engaged with the entire iterative machine learning spiral and design process. These various observational methods provide complementary insights into CE. Observing behavior and dialogue, as in Odgaard (Odgaard, 2023), captures dynamic, real-time learning processes, including sociocultural scaffolding and spontaneous problem-solving. Observing tangible outputs, as in Zimmermann-Niefield and colleagues (Zimmermann-Niefield et al., 2019) focuses on end results, revealing how students apply CE concepts creatively and critically. Together, these approaches provide a comprehensive view of both the CE process and the outcomes.

Interestingly, only 12 papers incorporated teachers' observations alongside researchers' analysis, showing limited integration of educators' perspectives in CE assessment. For instance, Bar-El and Worsley (Bar-El & Worsley, 2021) and Gourlet and Decortis (Gourlet & Decortis, 2018) incorporated teachers' observations to supplement their findings. Additionally, five papers referred to their observation method as "user testing" or "user studies", reflecting the influence of Human-Computer Interaction (HCI) terminology on CCI research. In line with the findings from the pior literature review on learning assessment in CCI (Baykal, Eriksson, & Torgersson, 2023), these studies evaluated learning results to validate their proposed technology prototypes or learning models. For example, Lunding and colleagues (Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022) assessed their prototype, ExposAR, by observing children's game designs and incorporating teachers' feedback, blending user-centered evaluation with educational assessment. While these terms "user testing and "user studies" are well-established within HCI, their use in the context of assessing CE raises questions about how researchers conceptualize children and their learning and empowerment, more than merely seeing them as users of the technology.

4.3.1.2. Feedback and reflection (n = 8, 15 %). Feedback and reflection are featured as assessment methods in eight papers, including peer feedback (n = 5), self-feedback (n = 3) and teacher feedback (n = 2). Peer evaluation encourages children to critically evaluate others' ideas, facilitating improvement (e.g., Gourlet, Eveillard, & Dervieux, 2016; Gourlet & Decortis, 2018; Ventä-Olkkonen et al., 2022). Self-feedback methods, such as reflective diaries as homework (e.g., Mahboob Kanafi, Iivari, Kinnula, et al., 2022), or performance rubrics for students to reflect on their understanding of data literacy to support learners' reflection on progress (e.g., Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022). Studies also included teacher feedback, often embedded in activities to motivate students and improve learning outcomes. For instance, Gourlet and Decortis (Gourlet & Decortis, 2018) used "a project board" as a collective critique tool, where students presented their work and received feedback from both teachers and peers. Students were reported being motivated by this process and took feedback seriously, fostering students' motivation and improvements in their projects.

4.3.1.3. Self-reports (n = 17, 32 %). Self-reports as assessment methods are featured in 17 papers, including interviews (n = 13) and questionnaires (n = 7). These methods typically evaluate students' attitudes toward or knowledge of the intended CE topics after interventions, as summative assessment. For example, Schaper and colleagues (Schaper et al., 2022) conducted interviews immediately after the intervention and one week later, exploring students' perceptions of activities and

their learning outcomes. Similarly, Lee and colleagues (Lee et al., 2023) employed identical questionnaires before and after their proposed exhibition to assess AI literacy competencies. These questionnaires examined participants' perceptions of ethical AI, its social impact, and the relationship between AI and themselves. Noteworthy is Tamashiro and colleagues (Tamashiro et al., 2023) designed "formative assessment task", where students completed a template in groups explaining IoT functionality. While this self-report activity demonstrated students' knowledge of intended learning outcomes, the researchers did not explicitly analyze its impact on improving the learning experience.

4.3.2. Assessment frameworks

Many studies **(n=20/53, 38 %)** in the review **do not specify theoretical frameworks** for analyzing or assessing CE outcomes. These papers often outline theories used to guide research design but lack elaboration on the theoretical foundations supporting their assessment of children's learning outcomes or learning process. For example, Antle and colleagues (Antle et al., 2021) mentioned using a critical reflection framework to guide their development of learning activities yet provide no details on the theories informing their assessment methods. This indicates a lack of explicit use of learning assessment frameworks, even though many studies aim to validate the effectiveness of technologies, learning activity models, or other CE initiatives.

4.3.2.1. Maker, CT, or CE literature (n = 9, 17 %). Nine papers in our corpus draw on Maker, CT, or CE literature to develop assessment methods and analytics lenses. These studies typically aligned their observations with core aspects of CE, such as children's reflections on the proposed technologies and the narratives children developed around artifacts (Huang et al., 2024). Yet, the chosen assessment methods are not justified. Three papers adapted established instruments from CT literature. For instance, Richard and Giri (Richard & Giri, 2019) adapted a computing attitudes survey, previously validated by Ericson and McKlin (Ericson & McKlin, 2015) and used in related maker literacy research (Kafai et al., 2014). These adaptations suggest that CT assessment can be adapted to address CE, but this approach may not fully capture CE's broader scope, including empowerment and reflective analysis.

4.3.2.2. Multidisciplinary frameworks (n = 7, 13 %). Seven papers developed their assessment methods grounded in multidisciplinary theories that are relevant to their learning designs. For instance, Ventä-Olkkonen et al. (Ventä-Olkkonen et al., 2022) used an Empowerment Analytical Lens to address bullying through critical design and maker activities. Similarly, Dwivedi and colleagues (Dwivedi et al., 2024) applied Rokeach Value Survey to examine values reflected in children's designs during AI literacy activities. These approaches illustrate how researchers tailor their frameworks to the disciplines with which their CE activities associate.

4.3.2.3. HCI-based approaches (n = 5, 9 %). Five papers based their assessment methods on **HCI literature**, particularly user testing and design thinking methodologies. For instance, Bekker and colleagues (Bekker et al., 2015) used a research-though-design approach to explore reflective design-based learning processes. Yet, without grounding in learning theories, questions arise about the rigor of such assessments.

4.3.2.4. Sociocultural perspectives (n = 7, 13 %). Seven papers employed **sociocultural perspectives**, rooted in disciplines such as psychology and education to observe how social and cultural interaction mediate learning. For instance, Odgaard (Odgaard, 2023) employed dialogism theories by Bakhtin and Wertsch (Bakhtin, 1986; Wertsch, 1998), applying the concepts of "appropriation" (i.e., adapting knowledge to personal meaning) and "mastery" (i.e., using knowledge as a tool for conflict or resistance) to analyze CE in CT activities. Other studies, such as (de Paula, Burn, Noss, & Valente, 2018; Lundmark & Jonsson, 2021) adopted ethnographic methods to observe children's behavior and dialogue in specific social contexts.

4.3.2.5. Grounding in the learning sciences (n = 2, 4 %). Two papers incorporated frameworks from the learning sciences. One paper by Antle and colleagues (Antle et al., 2022) explicitly combined HCI and the learning sciences frameworks, adapting concepts of critical reflection to evaluate design ethics in technical design task. The other paper by Schaper and colleagues (Schaper et al., 2022) conducted a design-based research study to investigate students' learning about emerging technologies during a five-day intervention in a primary school. Their use of the learning sciences frameworks indicates a more deliberate integration of learning theories, although such examples remain rare in CE literature.

4.4. Connection between assessment and CE activities

4.4.1. CE activity characteristics

We analyze the activities in CE studies through the lens of formative assessment practices (Black & Wiliam, 1998) and find that CE activities and formative assessment share many common characteristics in learning contexts. Specifically, these activities are characterized by several recurring strategies, such as experiential engagement, collaboration, raising awareness, contextualization, and structured reflection. These characteristics embody the core aspects and values of CE and promote CE through interactive, hands-on, and contextual learning.

4.4.1.1. Experiential learning (n = 34, 64 %). The majority of papers (n = 34) employ experiential learning activities, allowing children to engage through hands-on activities, such as construction, collaboration and real-world applications. Among these, 18 papers focus on constructing objects through maker activities, design, and play. As Papert (Papert, 1980) noted, construction-based approaches deepen understanding by engaging learners physically or digitally in building objects or solution, which is integral to both maker activities and STEM learning. For example, Huang and colleagues (Huang et al., 2024) designed a five-day workshop where students engaged in STEAM learning by constructing with IoT components and developing stories. One child participant demonstrated their understanding by integrating IoT components (e.g., LED, sound sensor) into a project simulating real-life scenarios.

Experiential learning often intersects with collaboration, as hands-on activities naturally promote teamwork and shared problem-solving experiences. Here 12 papers emphasize collaboration as a critical approach to foster inclusion and engagement. According to formative assessment theory, peer collaboration allows learners to serve as valuable learning resources for one another, particularly in contexts where instructors may not meet individual needs (Wiliam, 2018). For instance, Richard and Giri (Richard & Giri, 2019) employed bidirectionally responsive making activities in the curriculum to promote CE and suggested that collaborative teamwork was a key factor in reducing complexity of programming systems, reinforcing children's understanding of conceptual and procedural knowledge, and fostering inclusion, agency, and engagement among diverse learners.

Another key characteristic in experiential learning is the connection to real-world contexts. 15 papers bridge the gap between abstract concepts and tangible experiences, enhancing relevance and learner motivation (Murphy, 1997). For example, Schaper and colleagues (Schaper et al., 2022) introduced Augmented Reality and Machine Learning technologies through hands-on activities, demonstrating their real-world applications and societal impact. This aligns with the "Closeness" principle in CCI research (Schaper et al., 2022) which sparks children's interest by relating abstract concepts to familiar, real-world examples.

4.4.1.2. Structured reflection (n = 29, 55 %). Structured reflection is another key component in reinforcing CE by encouraging critical analysis of technology's role in society. Here 29 papers incorporate scaffolding mechanisms, such as structured prompts, digital tools or pedagogical approaches to support reflective thinking. For example, Antle and colleagues (Antle et al., 2021) created a deck of "reflection cards" to scaffold the reflection process and foster dialogues around ethical implications of design decisions. This structured approach made reflection interactive and accessible, ultimately enhancing learning outcomes. Furthermore, Iivari and colleagues (Iivari et al., 2023b) incorporated peer evaluation as their pedagogical approach, encouraging children to provide feedback on each other's work. This feedback process deepened students' understanding through reflection and critical analysis, further consolidating their knowledge.

4.4.1.3. Discussion (n = 21, 40 %). Discussion is one key characteristic in CE activities (n = 21), include brainstorming sessions and group discussions. These activities facilitate social interaction and dialogue, encouraging learners to share ideas, debate viable solutions, and refine their understanding through multiple perspectives. The social component and dialogic learning process of these activities not only enhance critical thinking and creativity but also foster communication skills. Moreover, discussion allows learners to articulate their understanding of the learning topic, thereby revealing their knowledge gap. For example, Bilstrup and colleagues (Bilstrup, Kaspersen, & Petersen, 2020) developed a card-based design workshop where high school students reflected on ethical dilemmas while designing their machine learning (ML) applications. During their discussion, students debated the ethical challenges but struggled with technical depth as the provided tools provided insufficient support for learning technical knowledge.

4.4.1.4. Awareness-raising (n = 16, 30 %). Finally, awareness-raising is often an entry point in the sequences of activities, with 16 papers employing introductory explanations, to familiarize children with the subject matter. These introductions provide context, highlight the relevance of the subject matter, or explain the mechanisms to be explored during the learning activities. For instance, Ventä-Olkkonen and colleagues (Ventä-Olkkonen et al., 2022) address the topic of bullying through critical design and making. Their initial learning activities entailed raising children's awareness about bullying and technology, encouraging children to reflect on the relationship between their empathy and technology in bullying situations. This early-stage framing improves children's learning expectations and ensures they understand the purpose and significance of the subject matter. Consistent with the formative assessment principles, this approach clarifies learning intentions, which boosts children's learning motivation and efficacy (Wiliam, 2018).

These findings showcase that researchers attempt to foster CE in children not through isolated activities, but rather through a carefully curated blend of activities. These collective activities nurture critical thinking and foster a nuanced understanding of societal implications of technologies through hands-on, interactive and collaborative learning in a real-world context. The activity characteristics of CE, such as structured reflection, peer evaluation, and introductory explanations, aligned well with formative assessment principles highlighting learning evidence, interactive feedback and metacognition. The following section presents the findings of frameworks that inform the development of these activities.

4.4.2. CE activity development frameworks

The theoretical foundations of these activities predominantly draw from the **CE, CT and Design theories** (n = 31, 58 %). These frameworks, including maker theories, AI literacy, data literacy, design frameworks and participatory design approaches, reflect the roots of CCI in HCI and Design disciplines, interdisciplinary efforts of CE to empower children through computational creativity, hands-on learning and problem solving.

Additionally, 23 papers (43 %) used **learning theories or instructional design theories** to shape learning activities. Among these, the most used theory is constructionism (n = 14), which aligns with the strong emphasis on hands-on activities in CCI literature. The second commonly used theory is the sociocultural theory (n = 8), including collaborative learning, cooperative inquiry-based learning, situated learning, and social-emotional learning. Other frameworks encompass, ARCS (an instructional design model) and multimodal learning.

Among them, 12 papers (23 %) **integrate both learning theories and CT/CE/Design literature**. For instance, Mannila and Skog (Mannila & Skog, 2023) structured their using the double diamond design process (i.e., which includes discover, define, develop, deliver) to frame learning as a design process. To maximize learning opportunities, they incorporated a sociocultural approach from the learning sciences, emphasizing the influence of social and cultural contexts on child development. This dual framework led to an interactive, participatory learning experience that involved co-creation, brainstorming, discussion and collaboration, ultimately reinforcing both the design outcomes and the learning goals.

These theoretical frameworks supporting the learning contexts of CE are interdisciplinary and diverse, primarily relying on CE, CT, and Design literature. Yet, a smaller subset enriches these approaches by integrating learning theories into existing CCI frameworks, creating robust approaches aimed at both design outcomes and learning goals (Fig. 2).

5. Discussion

This section addresses RQ3 by examining the insights from current CE research that can inform the development of more sustainable and theoretically robust assessment frameworks, and by proposing a future research agenda. First, to promote sustainable adoption of CE in K-12 Education, we advocate for recentering participatory design approaches in CE, thereby bridging the gap between CCI research and educational practices. Moreover, to strengthen the theoretical foundation of CE assessment framework, we propose building on the established instructional design model (Biggs, So-kum Tang, & Kennedy, 2022; Wiggins & McTighe, 2008) to advance CE research in three directions: (1) aligning and advancing CE learning goals, (2) integrating learning theories into assessment design, and (3) developing activities grounded in formative assessment principles. To support this proposition, we highlight exemplary CE studies from our review that serve as potential pathways for advancing research in these directions.

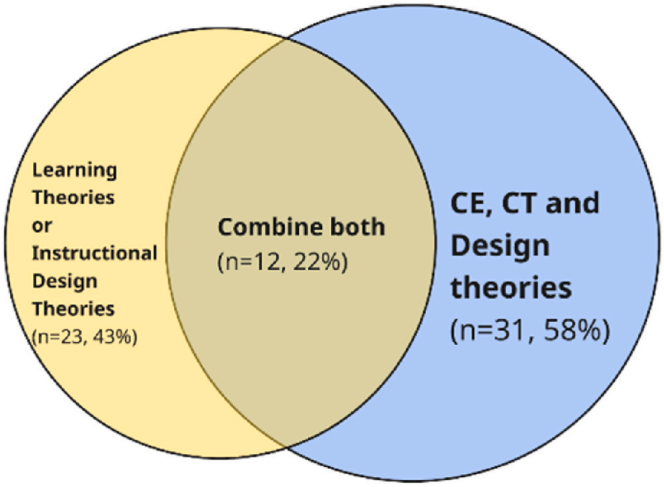


Fig. 2. Disciplines of frameworks adapted for developing CE activities.

5.1. Recentering participatory design approaches in CE

The notion of PD promotes active user participation in technology development, using ethnographic methods to unpack work practices and collaboratively, democratically shape future technology (Dindler et al., 2022). However, the findings in Section 4.1.1 reveal a notable absence of educators and teachers in the development of CE assessment methods and activities in educational settings which diverges from CE's roots in PD principles. Furthermore, the findings in Section 4.1.2 point out that few papers focus on understanding children's learning process and results. Most assessments documented in the reviewed corpus are conducted after workshops or interventions, indicating a reactive rather than proactive approach. This approach to assessment, typically conducted by researchers with minimal involvement from educators in the process, represents a missed opportunity for dynamic, real-time, formative feedback to promote learning and metacognition (Wafubwa & Csíkos, 2022; Wiliam, 2018).

Our findings show that positive outcomes are evident from the 13 papers that involved teachers in the CE assessment development process, highlighting the potential of collaboration with educators to enhance CE initiatives. For instance, Gourlet and Decortis (Gourlet & Decortis, 2018) collaborated with teachers to prototype learning artifacts that introduced making activities within classroom settings. This collaboration integrated these activities with teachers' existing pedagogical practices, allowing students to explore ways to demonstrate what they learned in an open-ended way. Gourlet and Decortis (Gourlet & Decortis, 2018) note that these making activities had a sustainable and progressive impact on teachers' existing practices. Similarly, Odgaard (Odgaard, 2023) involved educators in the observation and data analysis, illustrating how children can be empowered functionally, democratically, and critically empowered through learning. This empowerment was evidenced by excerpts of dialogue between children and educators, underscoring their distinct roles in fostering empowerment in various learning situations.

Given these positive CE outcomes from teacher involvement and the foundational role of PD in CE, we would like to encourage recentering PD methods in CE to prioritize collaboration with educators. Assessment practices in educational settings can reach their most sustainable potential when applied by teachers. Empowering educators to implement ongoing formative assessments during instruction could lead to more impactful learning outcomes, as they would be able to adjust their teaching strategies in response to students' evolving needs and progress (Dolin, Black, Harlen, & Tiberghien, 2018; Wiliam, 2018). This points to the value of stronger collaboration between CCI researchers and educational practitioners, to foster sustainable CE learning practices. Such collaboration ensures that assessment not only evaluates learning outcomes but also sustainably contributes to supporting educators' teaching confidence and enriching learners' learning experiences.

To recenter PD in CE and bridge the gap between CCI research and educators, creating a connection and common language between researchers and educator is essential. As such, we propose leveraging the established instructional design frameworks, such as constructive alignment (Biggs et al., 2022) and backward design (Wiggins & McTighe, 2008), which are widely recognized by educators. The instructional design framework highlights the alignment of learning goals, assessment, and activities, which offers a coherent structure for CE assessment. Learning goals should guide the design of learning activities and assessment. Learning activities, in turn, embed opportunities for assessment that are meaningful and reflective of the intended learning outcomes. Finally, learning assessment must be grounded in learning theories and generate learning evidence to iteratively adjust learning goals and activities, ensuring a continuous cycle of CE enhancement in K-12 education. Building on these insights, we propose the following future research directions.

5.2. Aligning and advancing CE learning goals

We advocate for aligning and advancing CE learning goals, since our findings show that current CE assessment practices have a critical gap of constructive alignment (Biggs, 1996). Section 4.3 reveals a critical gap that most assessment methods are not linked to specific learning goals, which inhibits rigorous evaluation of signs of learning. Section 4.2 indicates that the learning goals and progression levels for CE are non-specific and lacking concrete examples, which makes it difficult for curricular integration and limits the depth and impact of CE research (Mannila & Skog, 2023; Van Mechelen et al., 2023).

Nevertheless, based on six studies (Hatzigianni et al., 2021; Kaspersen et al., 2021; Iivari et al., 2023b; Mannila & Skog, 2023; Schaper et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022) in this literature review, we synthesized the themes and offer a pathway for refining and advancing current learning goals and progressions. Two primary categories for CE learning goals and progressions emerge: (1) progression across different age groups and (2) progression in the complexity of skills and depth of understanding. However, these models lack a clear definition of CE. For example, Kaspersen and colleagues (Kaspersen et al., 2021) acknowledge that the definition of CE remains somewhat ambiguous and hence CE is not fully represented in their proposed model. To address this limitation, this literature review identified core aspects of CE interpreted by CCI researchers. Our synthesized view indicates that the CE aspects converge around several core themes: critical reflection on the roles of technology, digital design literacy, digital literacy, and digital empowerment (agency). These aspects of CE can serve as pointers to refine the definition and scope of CE. Therefore, we suggest that researchers continue developing CE learning goals by integrating these identified CE themes into existing progression frameworks, such as the work of Kaspersen and colleagues (Kaspersen et al., 2021). Additionally, given that current CE learning goals remain abstract and better suited for curriculum-level or broader instructional units, it is also essential to translate these into specific, measurable, and observable learning objectives (Anderson, 2009) suited to individual learning units. Transforming CE themes into concrete learning goals and objectives provides a shared instructional language, bridging the gap and fostering collaboration between CCI and educators.

5.3. Integrating learning theories into CE assessments

Most studies in the review do not specify theoretical frameworks for CE assessment. If they do, they often ground their assessment approaches in CE literature and multidisciplinary theories related to design and HCI (See Section 4.3.2). This finding is consistent with observations of previous review studies (Baykal, Eriksson, & Torgersson, 2023; Van Mechelen et al., 2023). The absence of theoretical groundings in the learning sciences presents challenges to the consistency and validity of CE assessment practices. Without a robust theoretical foundation, it is difficult to determine whether chosen assessment methods adequately capture intended learning outcomes. This shortcoming might lead to inconsistent assessments and limit the practical application across diverse educational contexts.

To address this gap, we advocate for enhancing existing CE assessment practices by aligning them with established learning theories, thereby extracting more reliable learning evidence. We propose candidate learning theories to further the assessment methods, including Dialogic Learning Theory (Wegerif, 2007), Constructive Alignment (Biggs, 2005), Formative and Summative Assessment (Dolin et al., 2018), and Cultural-Historical Activity Theory (Grimalt-Álvaro & Ametller, 2021). In the following, we highlight studies from this review that exhibit potential link with these theories.

Several studies demonstrate diverse methods to embed feedback and reflection as assessment into activities, using mechanisms such as discussion, rubrics or diaries. These forms of embedded are closely aligned with formative assessment principles (Dolin et al., 2018). An illustrative

study is the work of Hatzigianni and colleagues (Hatzigianni et al., 2021), who provided a detailed account of observed actions, example dialogue, and project screenshots, systematically linking them to specific skills of digital design literacy. This approach exemplifies how observational assessment can be effectively used to connect learning evidence to intended learning outcomes, in accordance with the principles of Constructive Alignment (Biggs, 2005). Regarding self-reports methods, Vacca and colleagues (Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022) present a compelling approach by analyzing children's responses in interviews. They include excerpts that illustrate how children contextualize the data, articulate their relations to data, and reflect on the implications of data analysis. This approach resonates with dialogic learning theory (Wegerif, 2007), particularly in its emphasis on exploratory and meaning-making processes through dialogue.

While the above-mentioned studies provide valuable examples of assessment methods tailored to various research purposes, future research should seek to further refine these methods by explicitly adapting them to established learning theories and linking them to clearly defined learning goals.

5.4. Developing activities grounded in formative assessment principles

CE is aligned strongly with the concept behind formative assessment in terms of promoting agency, ownership, and metacognition. In this study, we noted that CE research efforts show unintentional alignment with formative assessment principles (See Section 4.4.1). Formative assessment requires not only assessment of learning outcomes, but also the application of learning evidence to adjust instruction and provide students with constructive feedback (Dolin, Black, Harlen, & Tiberghien, 2018; Wiliam, 2018). This feedback is crucial for fostering metacognitive abilities and intrinsic motivation (Wafubwa & Csíkos, 2022). Since formative assessments are embodied through learning experiences and embedded within learning activities, it is essential to enhance these activities through integration of formative assessment strategies and principles.

To achieve this enhancement, exemplary studies that show strong alignment with formative assessment principles offer a valuable foundation, particularly those developed through collaboration with educators. For example, Gourlet and Decortis (Gourlet & Decortis, 2018) collaborated with teachers to develop assessment and learning activities, aimed at fostering pupil engagement in making activities. They embedded assessment in activities through mechanisms such as discussion, peer and teacher feedback, and the use of project storyboards. The study explicitly defined the purpose of assessment activities as identifying pupils' skills, achievements, and areas for improvement, which aligned strongly with formative assessment principles. Moreover, they highlight the importance of incorporating mediational tools, such as storyboards, that enable pupils to document their activities. These tools allow pupils to reflect on the strategies they used to achieve their learning objectives, promoting metacognition. Their findings also revealed that these assessment designs transformed the classroom environment, promoting sustained learning experiences through making activities. These results demonstrate that integrating formative assessment principles into CE has the rich potential to transform CE research initiatives into more sustainable practices within educational contexts.

6. Conclusion

This review aims to move beyond the good intentions underpinning the proliferation of CE-related research efforts by examining current CE assessment practices and outlining a future research agenda to build a more robust and sustainable CE assessment framework. While at first glance the field may appear to have significant gaps in assessment practices, a closer examination reveals promising examples that can serve as a foundation for advancement. Our findings indicate that current CE assessment practices are primarily researcher-driven, focusing on evaluating the effectiveness of learning models. However, these efforts often lack constructive alignment between learning goals, activities, and assessment methods. The predominant assessment methods involve ongoing observation of dialogue, behavior, and project work, which, while valuable, remain limited in scope for empowering education practitioners. Despite the limitations, we identified exemplary assessment practices from the existing literature that serve as a starting point to develop a shared language between educators and CCI researchers. We advocate for a shift in CE assessment practices that centers frontline educators as key stakeholders. By cultivating sustainable and participatory collaborations with educators, the CCI community can develop approaches that are better integrated into real-world learning contexts and thus more effectively promote CE. To support this shift, we propose leveraging the good examples identified in this review to enhance the alignment of instructional design's three pillars — learning goals, activities, and assessment. We suggest synthesizing CE aspects into learning goals, integrating learning theories into assessment, and designing CE activities with formative assessment principles. These directions not only enrich CE assessment practices but may also hold broader relevance for skill-based approaches such as CT and design literacy—particularly when these fields seek to incorporate embedded formative strategies to foster reflective and critical engagement. These directions will help move our good intentions closer to our goal of empowering children in the digitized world.

CRediT authorship contribution statement

Yu-Yu Liu: Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christian Dindler:** Writing – review & editing, Supervision, Conceptualization. **Ole S. Iversen:** Writing – review & editing, Supervision, Conceptualization. **Yasmin B. Kafai:** Writing – review & editing.

Selection and participation of children

No children participated in this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We sincerely thank our colleagues and the anonymous reviewers for valuable feedback. This work was supported by a research grant (NNF23SA0082894) from Novo Nordisk Foundation, a research grant (#54607) from VILLUM FONDEN, and a research grant (R419-2022-1261) from Lundbeck Foundation.

A APPENDIX

A.1 Traceability Table of Reviewed Studies by Finding Theme

Finding Category	Sub-Theme	Number of Papers (n)	Study IDs/References
4.1.1			
Stakeholder roles	Researcher-led	25	(Antle et al., 2021; Antle et al., 2022; Braswell et al., 2021; Brooks & Sjöberg, 2020; Druga et al., 2022; Dwivedi et al., 2024; Richard & Giri, 2019; Gourlet et al., 2016; Kaspersen et al., 2021; Huang et al., 2024; Kaspersen et al., 2021; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Kong et al., 2019; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Maqsood et al., 2018; Schaper et al., 2023; Szczuka et al., 2022; Vacca, DesPortes, Tes, Silander, Amato, et al., 2022; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Xiang et al., 2023; Zimmermann-Niefield et al., 2019)
	Teacher involved	13	(Tamashiro et al., 2023; Odgaard, 2023; Bekker et al., 2015; Chu et al., 2017; Dindler et al., 2023; Göbl et al., 2023; Gourlet & Decortis, 2018; Hatzigianni et al., 2021; de Paula, Burn, Noss, & Valente, 2018; Schaper et al., 2022; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022)
	Student involved	6	(Chatlani et al., 2023; Coenraad, 2024; Dowthwaite et al., 2020; Gourlet & Decortis, 2018; Morales-Navarro, Thompson, et al., 2022; Yap & Lee, 2020)
	Designer/artist involved	3	(Bustos et al., 2024; Giannakos & Jaccheri, 2018; Lee et al., 2023)
	Multiple stakeholders	1	Bekker et al. (2015)
	Teacher-led without researcher involvement	1	Van Mechelen, Wagner, et al. (2021)
Learning context	Formal (schools)	29	(Tamashiro et al., 2023; Odgaard, 2023; Bekker et al., 2015; Börjesson et al., 2019; Brooks & Sjöberg, 2020; Chu et al., 2017; Dindler et al., 2023; Eriksson et al., 2019; Richard & Giri, 2019; Göbl et al., 2023; Gourlet & Decortis, 2018; Hatzigianni et al., 2021; Hjorth et al., 2016; Kaspersen et al., 2021; Kaspersen et al., 2021; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Kong et al., 2019; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Schaper et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Vacca, DesPortes, Tes, Silander, Amato, et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Van Mechelen, Wagner, et al., 2021; Ventä-Olkkonen et al., 2022; Xiang et al., 2023)
	Non-formal (afterschool, labs)	14	(Antle et al., 2021, 2022; Braswell et al., 2021; Bustos et al., 2024; Chatlani et al., 2023; Dowthwaite et al., 2020; Dwivedi et al., 2024; Giannakos & Jaccheri, 2018; de Paula, Burn, Noss, & Valente, 2018; Huang et al., 2024; Lundmark & Jonsson, 2021; Morales-Navarro, Thompson, et al., 2022; Sosa et al., 2022; Weibert et al., 2017)
	Both formal and non-formal learning contexts	4	(Brooks & Sjöberg, 2020; Coenraad, 2024; Gourlet et al., 2016; Morales-Navarro, Kafai, & Gregory, 2022)
	Informal (home, unstructured)	4	(Druga et al., 2022; Lee et al., 2023; Szczuka et al., 2022; Yap & Lee, 2020)
	All applicable	2	(Kafai, Proctor, & Lui, 2019; Zimmermann-Niefield et al., 2019)
Educational level	Grades 1–6	26	(Bekker et al., 2015; Börjesson et al., 2019; Braswell et al., 2021; Brooks & Sjöberg, 2020; Chu et al., 2017; Coenraad, 2024; Dindler et al., 2023; Dowthwaite et al., 2020; Druga et al., 2022; Dwivedi et al., 2024; Giannakos & Jaccheri, 2018; Gourlet & Decortis, 2018; Gourlet et al., 2016; Hatzigianni et al., 2021; Huang et al., 2024; Iivari et al., 2023b; Kong et al., 2019; Mannila & Skog, 2023; Lundmark & Jonsson, 2021; Maqsood et al., 2018; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Szczuka et al., 2022; Van Mechelen, Wagner, et al., 2021; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Yap & Lee, 2020)
	Grades 7-9	21	(Tamashiro et al., 2023; Odgaard, 2023; Antle et al., 2021; Antle et al., 2022; Bekker et al., 2015; Börjesson et al., 2019; Braswell et al., 2021; Brooks & Sjöberg, 2020; Chatlani et al., 2023; Chu et al., 2017; Dindler et al., 2023; Dowthwaite et al., 2020; Eriksson et al., 2019; Richard & Giri, 2019; Göbl et al., 2023; Gourlet & Decortis, 2018; Hatzigianni et al., 2021; de Paula, Burn, Noss, & Valente, 2018; Hjorth et al., 2016; Kaspersen et al., 2021; Kaspersen et al., 2021; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Kong et al., 2019; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Morales-Navarro, Kafai, & Gregory, 2022; Morales-Navarro, Thompson, et al., 2022; Schaper et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Amato, et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Van Mechelen, Wagner, et al., 2021; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Xiang et al., 2023; Yap & Lee, 2020)
	Grades 10-12	11	(Bustos et al., 2024; Chatlani et al., 2023; Dowthwaite et al., 2020; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Kaspersen et al., 2021; Kaspersen et al., 2022; Bilstrup & Kaspersen, 2020; Morales-Navarro, Kafai, & Gregory, 2022; Sosa et al., 2022; Yap & Lee, 2020)
	Grades K-12	6	(Eriksson et al., 2019; Göbl et al., 2023; Hjorth et al., 2016; Kaspersen et al., 2021; Kafai, Proctor, & Lui, 2019; Zimmermann-Niefield et al., 2019)
	Kindergarten	4	(Odgaard, 2023; Börjesson et al., 2019; Hatzigianni et al., 2021; Xiang et al., 2023)
	Youth	1	Lee et al. (2023)
Geographical Location	USA	13	(Bustos et al., 2024; Chatlani et al., 2023; Chu et al., 2017; Coenraad, 2024; Druga et al., 2022; Dwivedi et al., 2024; Richard & Giri, 2019; Huang et al., 2024; Kafai, Proctor, & Lui, 2019; Morales-Navarro, Thompson, et al., 2022; Vacca et al., 2022a, 2022b; Zimmermann-Niefield et al., 2019)
	Canada	3	(Antle et al., 2021, 2022; Maqsood et al., 2018)
	Both USA and Canada	1	Braswell et al. (2021)
	Europe	30	(Tamashiro et al., 2023; Odgaard, 2023; Bekker et al., 2015; Börjesson et al., 2019; Brooks & Sjöberg, 2020; Dindler et al., 2023; Dowthwaite et al., 2020; Eriksson et al., 2019; Giannakos & Jaccheri, 2018; Göbl et al., 2023; Gourlet & Decortis, 2018; Gourlet et al., 2016; Hatzigianni et al., 2021; de Paula, Burn, Noss, & Valente, 2018; Hjorth et al., 2016; Kaspersen et al., 2021; Kaspersen et al., 2021; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Mannila & Skog, 2023; Lundmark & Jonsson, 2021; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Schaper et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, &

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Finding Category	Sub-Theme	Number of Papers (n)	Study IDs/References
			Petersen, 2022; Szczuka et al., 2022; Van Mechelen, Wagner, et al., 2021; Ventä-Olkkonen et al., 2022; Weibert et al., 2017)
	Denmark	13	(Tamashiro et al., 2023; Odgaard, 2023; Dindler et al., 2023; Eriksson et al., 2019; Hjorth et al., 2016; Kaspersen et al., 2021; Kaspersen et al., 2021; Kaspersen et al., 2022; Bilstrup & Kaspersen, 2020; Schaper et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Van Mechelen, Wagner, et al., 2021)
	Finland	4	(Iivari et al., 2023b; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Ventä-Olkkonen et al., 2022)
	Sweden	3	(Börjesson et al., 2019; Brooks & Sjöberg, 2020; Lundmark & Jonsson, 2021)
	Asia–Pacific (China, Korea, New Zealand, Singapore)	5	(Kong et al., 2019; Lee et al., 2023; Sosa et al., 2022; Xiang et al., 2023; Yap & Lee, 2020)
	Global	1	Morales-Navarro, Kafai, and Gregory (2022)
4.1.2			
Purpose	Evaluating the effectiveness of learning activities/models	19	(Antle et al., 2021; Antle et al., 2022; Braswell et al., 2021; Brooks & Sjöberg, 2020; Dowthwaite et al., 2020; Druga et al., 2022; Dwivedi et al., 2024; Giannakos & Jaccheri, 2018; Göbl et al., 2023; Hatzigianni et al., 2021; Huang et al., 2024; Kaspersen et al., 2022; Bilstrup & Kaspersen, 2020; Lee et al., 2023; Mannila & Skog, 2023; Morales-Navarro, Thompson, et al., 2022; Sosa et al., 2022; Weibert et al., 2017; Xiang et al., 2023)
	Assessing technological designs for CE	10	(Tamashiro et al., 2023; Antle et al., 2021; Gourlet et al., 2016; de Paula, Burn, Noss, & Valente, 2018; Kaspersen et al., 2021; Kaspersen et al., 2022; Maqsood et al., 2018; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Yap & Lee, 2020; Zimmermann-Niefield et al., 2019)
	Validating proposed CE frameworks	9	(Bekker et al., 2015; Bustos et al., 2024; Richard & Giri, 2019; Hjorth et al., 2016; Kaspersen et al., 2021; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Schaper et al., 2022, 2023; Ventä-Olkkonen et al., 2022)
	Deepening understanding of CE learning processes	9	(Odgaard, 2023; Chu et al., 2017; Gourlet & Decortis, 2018; Hatzigianni et al., 2021; Iivari et al., 2023b; Schaper et al., 2023; Szczuka et al., 2022; Vacca et al., 2022a, 2022b)
	Research through design	4	(Druga et al., 2022; Kong et al., 2019; Lundmark & Jonsson, 2021; Yap & Lee, 2020)
4.2			
Aspects of CE being assessed	Critical reflection	27	(Tamashiro et al., 2023; Odgaard, 2023; Antle et al., 2021; Antle et al., 2022; Braswell et al., 2021; Coenraad, 2024; Dowthwaite et al., 2020; Eriksson et al., 2019; Gourlet & Decortis, 2018; Gourlet et al., 2016; Kaspersen et al., 2021; Huang et al., 2024; Kaspersen et al., 2022; Iivari et al., 2023b; Kafai, Proctor, & Lui, 2019; Bilstrup & Kaspersen, 2020; Lee et al., 2023; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Morales-Navarro, Kafai, & Gregory, 2022; Morales-Navarro, Thompson, et al., 2022; Schaper et al., 2022; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Amato, et al., 2022; Ventä-Olkkonen et al., 2022; Weibert et al., 2017)
	Digital literacy	18	(Antle et al., 2022; Bekker et al., 2015; Chatlani et al., 2023; Dowthwaite et al., 2020; Druga et al., 2022; Dwivedi et al., 2024; Hatzigianni et al., 2021; Kaspersen et al., 2021; Lee et al., 2023; Maqsood et al., 2018; Schaper et al., 2022, 2023; Szczuka et al., 2022; Vacca et al., 2022a, 2022b; Van Mechelen, Wagner, et al., 2021; Weibert et al., 2017; Yap & Lee, 2020)
	Digital empowerment	18	(Odgaard, 2023; Antle et al., 2022; Bustos et al., 2024; Dindler et al., 2023; Göbl et al., 2023; Gourlet & Decortis, 2018; Kaspersen et al., 2021; Iivari et al., 2023b; Kong et al., 2019; Lundmark & Jonsson, 2021; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Schaper et al., 2022, 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Van Mechelen, Wagner, et al., 2021; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Yap & Lee, 2020)
	Digital design literacy	19	(Bekker et al., 2015; Börjesson et al., 2019; Brooks & Sjöberg, 2020; Chu et al., 2017; Eriksson et al., 2019; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Göbl et al., 2023; Gourlet & Decortis, 2018; Hatzigianni et al., 2021; de Paula, Burn, Noss, & Valente, 2018; Hjorth et al., 2016; Bilstrup & Kaspersen, 2020; Lundmark & Jonsson, 2021; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Morales-Navarro, Thompson, et al., 2022; Van Mechelen, Wagner, et al., 2021; Xiang et al., 2023; Zimmermann-Niefield et al., 2019)
4.2.1			
Learning goals	Do not specify learning goals	26	(Odgaard, 2023; Antle et al., 2021; Brooks & Sjöberg, 2020; Bustos et al., 2024; Dwivedi et al., 2024; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Gourlet & Decortis, 2018; Gourlet et al., 2016; de Paula, Burn, Noss, & Valente, 2018; Hjorth et al., 2016; Huang et al., 2024; Kaspersen et al., 2022; Bilstrup & Kaspersen, 2020; Kong et al., 2019; Lundmark & Jonsson, 2021; Maqsood et al., 2018; Morales-Navarro, Thompson, et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Szczuka et al., 2022; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Xiang et al., 2023; Zimmermann-Niefield et al., 2019)
	Outline learning goals	18	(Tamashiro et al., 2023; Antle et al., 2022; Bekker et al., 2015; Börjesson et al., 2019; Braswell et al., 2021; Chu et al., 2017; Dowthwaite et al., 2020; Druga et al., 2022; Hatzigianni et al., 2021; Kaspersen et al., 2021; Iivari et al., 2023b; Kafai, Proctor, & Lui, 2019; Lee et al., 2023; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Schaper et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Yap & Lee, 2020)
Progression in CE skills	Progression among different ages	3	(Hatzigianni et al., 2021; Iivari et al., 2023b; Mannila & Skog, 2023)
	Progression in the complexity of skills and understanding	3	(Kaspersen et al., 2021; Schaper et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022)
4.3.1			
Assessment methods	Observation	41	(Tamashiro et al., 2023; Odgaard, 2023; Antle et al., 2021; Antle et al., 2022; Bekker et al., 2015; Börjesson et al., 2019; Brooks & Sjöberg, 2020; Chu et al., 2017; Dowthwaite et al., 2020; Druga et al., 2022; Dwivedi et al., 2024; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Göbl et al., 2023; Gourlet & Decortis, 2018; Gourlet et al., 2016; Hatzigianni et al., 2021; de

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Finding Category	Sub-Theme	Number of Papers (n)	Study IDs/References
			Paula, Burn, Noss, & Valente, 2018; Hjorth et al., 2016; Kaspersen et al., 2021; Huang et al., 2024; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Mannila & Skog, 2023; Lundmark & Jonsson, 2021; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Maqsood et al., 2018; Morales-Navarro, Thompson, et al., 2022; Schaper et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Amato, et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Van Mechelen, Wagner, et al., 2021; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Xiang et al., 2023; Yap & Lee, 2020; Zimmermann-Niefield et al., 2019)
	Feedback and reflection	8	(Gourlet & Decortis, 2018; Gourlet et al., 2016; Iivari et al., 2023b; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Van Mechelen, Wagner, et al., 2021; Ventä-Olkkonen et al., 2022)
	Self-reports	17	(Tamashiro et al., 2023; Braswell et al., 2021; Bustos et al., 2024; Dowthwaite et al., 2020; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Iivari et al., 2023b; Kong et al., 2019; Lee et al., 2023; Lundmark & Jonsson, 2021; Maqsood et al., 2018; Schaper et al., 2022, 2023; Szczuka et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Ventä-Olkkonen et al., 2022; Yap & Lee, 2020)
4.3.2			
Assessment frameworks	Do not specify theoretical frameworks	20	(Tamashiro et al., 2023; Antle et al., 2021; Börjesson et al., 2019; Chu et al., 2017; Dowthwaite et al., 2020; Göbl et al., 2023; Hjorth et al., 2016; Kaspersen et al., 2021; Kaspersen et al., 2022; Bilstrup & Kaspersen, 2020; Lee et al., 2023; Mannila & Skog, 2023; Morales-Navarro, Thompson, et al., 2022; Schaper et al., 2023; Szczuka et al., 2022; Vacca, DesPortes, Tes, Silander, Amato, et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Van Mechelen, Wagner, et al., 2021; Yap & Lee, 2020; Zimmermann-Niefield et al., 2019)
	Marker, CT, or CE literature	9	(Braswell et al., 2021; Bustos et al., 2024; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Huang et al., 2024; Kong et al., 2019; Lundmark & Jonsson, 2021; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Xiang et al., 2023)
	Multidisciplinary frameworks	7	(Dwivedi et al., 2024; Giannakos & Jaccheri, 2018; Iivari et al., 2023b; Kong et al., 2019; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Ventä-Olkkonen et al., 2022; Weibert et al., 2017)
	HCI-based approaches	5	(Antle et al., 2022; Bekker et al., 2015; Gourlet et al., 2016; Hatzigianni et al., 2021; Maqsood et al., 2018)
	Sociocultural perspectives	7	(Odgaard, 2023; Brooks & Sjöberg, 2020; Druga et al., 2022; Gourlet & Decortis, 2018; de Paula, Burn, Noss, & Valente, 2018; Lundmark & Jonsson, 2021; Sosa et al., 2022)
	Learning sciences	2	(Antle et al., 2022; Schaper et al., 2022)
4.4.1			
CE activity characteristics	Experiential learning	34	(Tamashiro et al., 2023; Odgaard, 2023; Antle et al., 2022; Bekker et al., 2015; Braswell et al., 2021; Brooks & Sjöberg, 2020; Bustos et al., 2024; Chatlani et al., 2023; Chu et al., 2017; Coenraad, 2024; Dindler et al., 2023; Druga et al., 2022; Dwivedi et al., 2024; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Göbl et al., 2023; Hatzigianni et al., 2021; Hjorth et al., 2016; Kaspersen et al., 2021; Huang et al., 2024; Iivari et al., 2023b; Bilstrup, Kaspersen, & Petersen, 2020; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Maqsood et al., 2018; Morales-Navarro, Kafai, & Gregory, 2022; Schaper et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Vacca, DesPortes, Tes, Silander, Amato, et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Xiang et al., 2023)
	Structured reflection	29	(Odgaard, 2023; Antle et al., 2021; Antle et al., 2022; Bekker et al., 2015; Brooks & Sjöberg, 2020; Chu et al., 2017; Dindler et al., 2023; Druga et al., 2022; Dwivedi et al., 2024; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Göbl et al., 2023; Gourlet & Decortis, 2018; Gourlet et al., 2016; Hjorth et al., 2016; Kaspersen et al., 2021; Huang et al., 2024; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Lee et al., 2023; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Morales-Navarro, Thompson, et al., 2022; Schaper et al., 2023; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Yap & Lee, 2020)
	Discussion	21	(Tamashiro et al., 2023; Antle et al., 2022; Bekker et al., 2015; Braswell et al., 2021; Coenraad, 2024; Dindler et al., 2023; Dowthwaite et al., 2020; Dwivedi et al., 2024; Kaspersen et al., 2021; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Morales-Navarro, Kafai, & Gregory, 2022; Morales-Navarro, Thompson, et al., 2022; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Weibert et al., 2017; Yap & Lee, 2020)
	Awareness-raising	16	(Tamashiro et al., 2023; Antle et al., 2021; Bustos et al., 2024; Coenraad, 2024; Dowthwaite et al., 2020; Richard & Giri, 2019; de Paula, Burn, Noss, & Valente, 2018; Huang et al., 2024; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Mannila & Skog, 2023; Schaper et al., 2022; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Ventä-Olkkonen et al., 2022; Yap & Lee, 2020)
4.4.2			
CE activity development framework	CE, CT and design theories	31	(Tamashiro et al., 2023; Odgaard, 2023; Antle et al., 2021; Antle et al., 2022; Brooks & Sjöberg, 2020; Coenraad, 2024; Dindler et al., 2023; Dowthwaite et al., 2020; Druga et al., 2022; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Göbl et al., 2023; Gourlet & Decortis, 2018; Hatzigianni et al., 2021; Hjorth et al., 2016; Huang et al., 2024; Kaspersen et al., 2022; Iivari et al., 2023b; Bilstrup & Kaspersen, 2020; Lee et al., 2023; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Maqsood et al., 2018; Morales-Navarro, Kafai, & Gregory, 2022; Schaper et al., 2022; Schaper et al., 2023; Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022; Sosa et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Weibert et al., 2017; Xiang et al., 2023)

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Finding Category	Sub-Theme	Number of Papers (n)	Study IDs/References
	Learning/instructional design theories	23	(Antle et al., 2022; Bekker et al., 2015; Braswell et al., 2021; Brooks & Sjöberg, 2020; Bustos et al., 2024; Dwivedi et al., 2024; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Gourlet et al., 2016; Hatzigianni et al., 2021; de Paula, Burn, Noss, & Valente, 2018; Kaspersen et al., 2021; Kaspersen et al., 2022; Kafai, Proctor, & Lui, 2019; Bilstrup & Kaspersen, 2020; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Szczuka et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Ventä-Olkkonen et al., 2022; Weibert et al., 2017; Xiang et al., 2023; Yap & Lee, 2020)
	Both learning theories and CT/CE/ Design literature	12	(Antle et al., 2022; Brooks & Sjöberg, 2020; Richard & Giri, 2019; Giannakos & Jaccheri, 2018; Hatzigianni et al., 2021; Kaspersen et al., 2022; Bilstrup & Kaspersen, 2020; Mannila & Skog, 2023; Mahboob Kanafi, Iivari, Kinnula, et al., 2022; Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022; Weibert et al., 2017; Xiang et al., 2023)

A.2 Overview of scoped corpus and their assessment methods

	Observation Behaviors (n = 29) Dialogues (n = 28) Project work (n = 28) Teacher observation (n = 12) User testing (n = 5)	Feedback and reflection Peer feedback (n = 5) Self-feedback (n = 4) Teacher feedback (n = 3)	Self-reports Interview (n = 13) Questionnaire (n = 7) Mid-way knowledge check (n = 1)	No assessment
1: (Yap & Lee, 2020)	Behaviors, user testing	–	Interviews	–
2: (Maqsood et al., 2018)	Behaviors, dialogues, user testing	–	Interviews	–
3: (Brooks & Sjöberg, 2020)	Behaviors, dialogues, project work	–	–	–
4: (Ventä-Olkkonen et al., 2022)	Project work	Peer feedback	Interviews	–
5: (Lundmark & Jonsson, 2021)	Behaviors, dialogues, project work	–	Interviews	–
6: (Chu et al., 2017)	Behaviors, dialogues, project work	–	–	–
7: (Schaper et al., 2022)	Teacher observation	–	Interviews	–
8: (Iivari et al., 2023b)	Behaviors and project work	Peer feedback	Interviews	–
9: (Weibert et al., 2017)	Behaviors, dialogues, project work, and teacher observation	–	–	–
10: (Huang et al., 2024)	Behaviors, dialogues, project work	–	–	–
11: (Kong et al., 2019)	–	–	Questionnaire	–
12: (Richard & Giri, 2019)	Behaviors	–	Interviews	–
13: (Van Mechelen, Wagner, et al., 2021)	Behaviors, dialogues, teacher observation	Teacher feedback and self-feedback	–	–
14: (Dindler et al., 2023)	–	–	–	No assessment applied to learners
15: (Hjorth et al., 2016)	Behaviors, project work,	–	–	–
16: (Dwivedi et al., 2024)	Behaviors, dialogues, project work, User testing	–	–	–
17: (Lunding, Grønbæk, Bilstrup, Sørensen, & Petersen, 2022)	Behaviors, dialogues, project work, user testing	–	–	–
18: (Druga et al., 2022)	Behaviors, dialogues, project work	–	–	–
19: (Schaper et al., 2023)	Dialogues	–	Interviews	–
20: (Sosa et al., 2022)	Dialogues	Peer feedback	–	–
21: (Lee et al., 2023)	–	–	Interviews and questionnaires	–
22: (Giannakos & Jaccheri, 2018)	Dialogues	–	Interviews and questionnaires	–
23: (Kafai, Proctor, & Lui, 2019)	–	–	–	No assessment applied to learners
24: (Kaspersen et al., 2022)	Behaviors, dialogues, project work	–	–	–
25: (Szczuka et al., 2022)	–	–	Interviews	–
26: (Bustos et al., 2024)	–	–	Interviews and questionnaires	–
27:(Morales-Navarro et al., 2022)	–	–	–	No assessment of learners' learning of CE- only analysis of learning activities
28: (Dowthwaite et al., 2020)	Dialogues	–	Questionnaires	–
29: (Xiang et al., 2023)	Teacher observation	–	–	–
30: (Coenraad, 2024)	–	–	–	No assessment applied to learners
31: (Antle et al., 2021)	Behaviors, dialogues, project work	–	–	–
32: (Braswell et al., 2021)	–	–	Questionnaires	–
33: (de Paula, Burn, Noss, & Valente, 2018)	Behaviors, dialogues, project work	–	–	–
34: (Gourlet & Decortis, 2018)	Behaviors, dialogues, project work, teacher observation	Peer feedback, self-feedback, teacher feedback	–	–
35: (Morales-Navarro, Thompson, et al., 2022)	Behaviors, dialogues, project work,	–	–	–
36: (Göbl et al., 2023)	Teacher observation	–	–	–
37: (Zimmermann-Niefield et al., 2019)	Behaviors, project work	–	–	–
38: (Bilstrup, Kaspersen & Petersen, 2020)	Behaviors, dialogues, project work,	–	–	–
39: (Börjesson et al., 2019)	Teacher observation	–	–	–

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	Observation Behaviors (n = 29) Dialogues (n = 28) Project work (n = 28) Teacher observation (n = 12) User testing (n = 5)	Feedback and reflection Peer feedback (n = 5) Self-feedback (n = 4) Teacher feedback (n = 3)	Self-reports Interview (n = 13) Questionnaire (n = 7) Mid-way knowledge check (n = 1)	No assessment
40: (Bekker et al., 2015)	Behaviors, dialogues, project work, teacher observation, user testing	–	–	–
41: (Tamashiro et al., 2023)	Behaviors, dialogues, project work, teacher observation	–	Midway knowledge check	–
42: (Chatlani et al., 2023)	–	–	–	No assessment applied to learners
43: (Gourlet et al., 2016)	Behaviors, dialogues, project work	Peer feedback	–	–
44: (Kaspersen, Graungaard, Bouvin, Petersen, & Eriksson, 2021)	–	–	–	No assessment applied to learners
45: (Mahboob Kanafi et al., 2022)	Behaviors, dialogues, project work	Self-feedback	–	–
46: (Kaspersen et al., 2021)	Behaviors, dialogues, project work	–	–	–
47: (Vacca, DesPortes, Tes, Silander, Amato, et al., 2022)	Project work	–	–	–
48: (Odgaard, 2023)	Behaviors, dialogues, teacher observation	–	–	–
49: (Eriksson et al., 2019)	–	–	–	No assessment applied to learners
50: (Hatzigianni et al., 2021)	Behaviors, dialogues, project work	–	–	–
51: (Mannila & Skog, 2023)	Project work, teacher observation	–	–	–
52: (Antle et al., 2022)	Behaviors, dialogues, project work	–	–	–
53: (Vacca, DesPortes, Tes, Silander, Matuk, et al., 2022)	Project work, teacher observation	Self-feedback	Interviews	–

Data availability

Data will be made available on request.

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